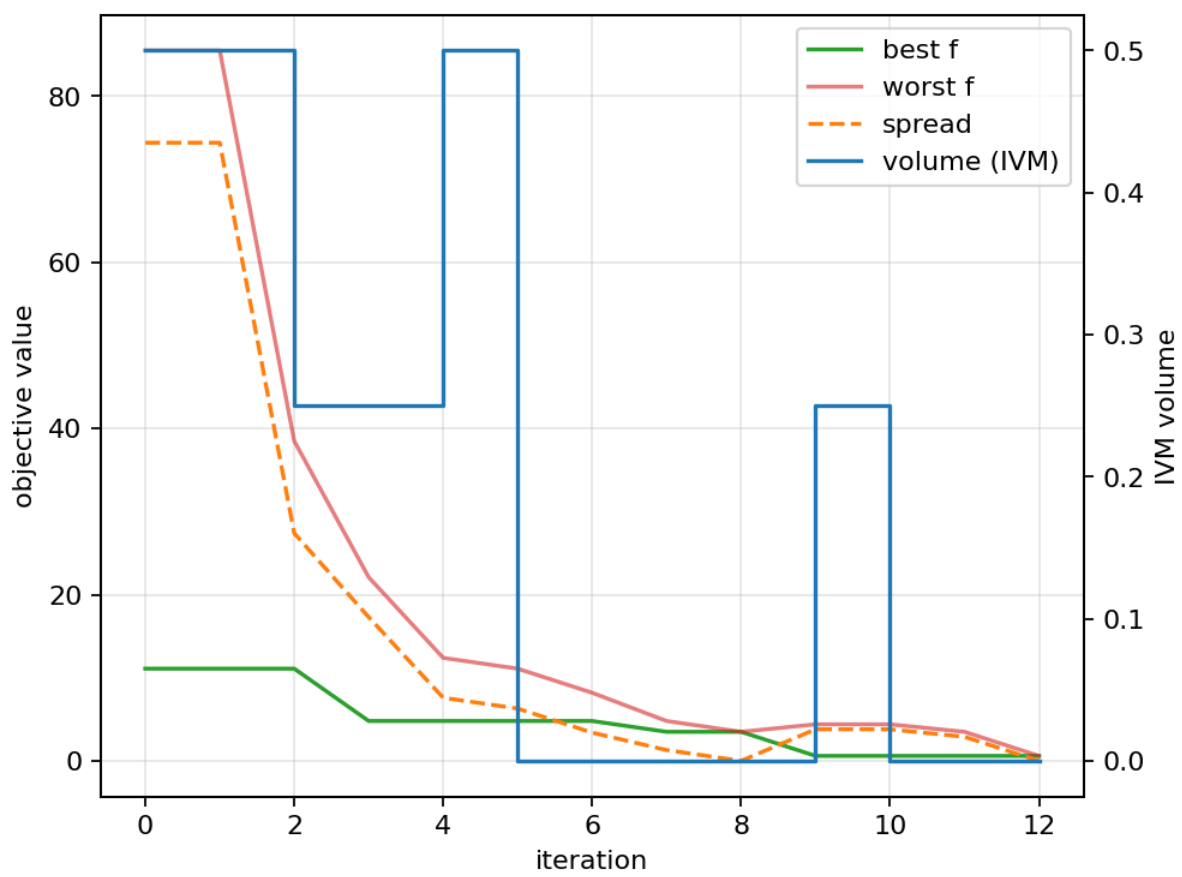
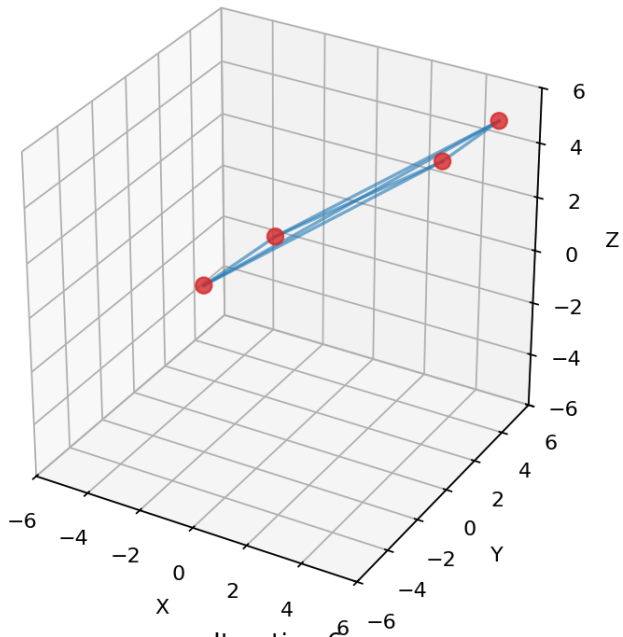


$\alpha \approx 1$
 $\gamma \approx 2$
 $\rho \approx 0.5$
 $\sigma \approx 0.5$

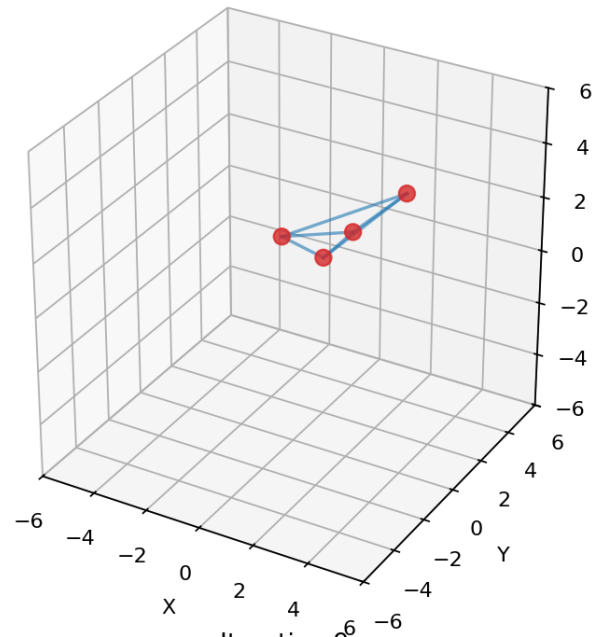


Nelder-Mead Simplex Evolution on Integer Quadray Lattice

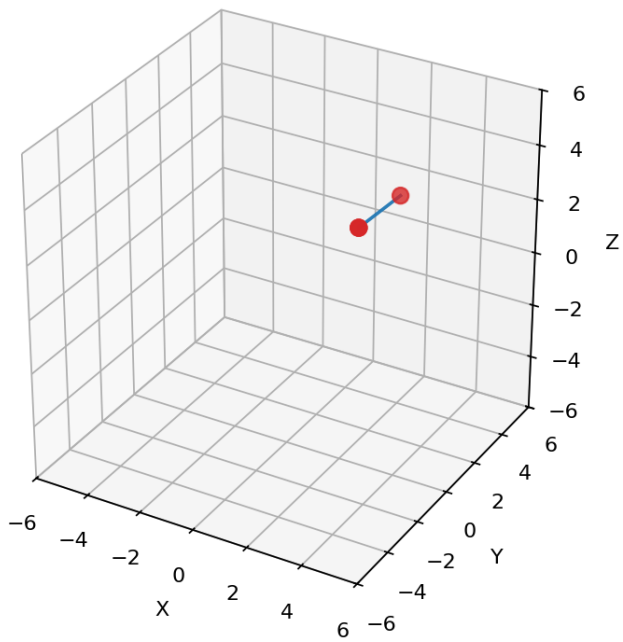
Iteration 0
Best: 11.1, Spread: 74.4



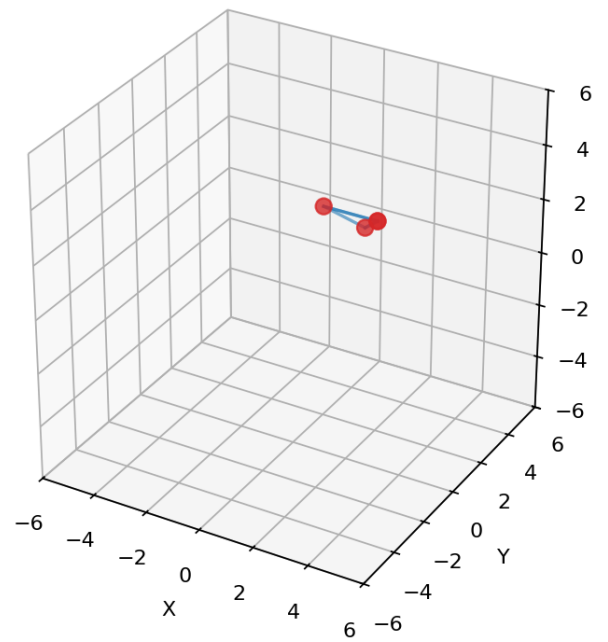
Iteration 3
Best: 4.8, Spread: 7.6



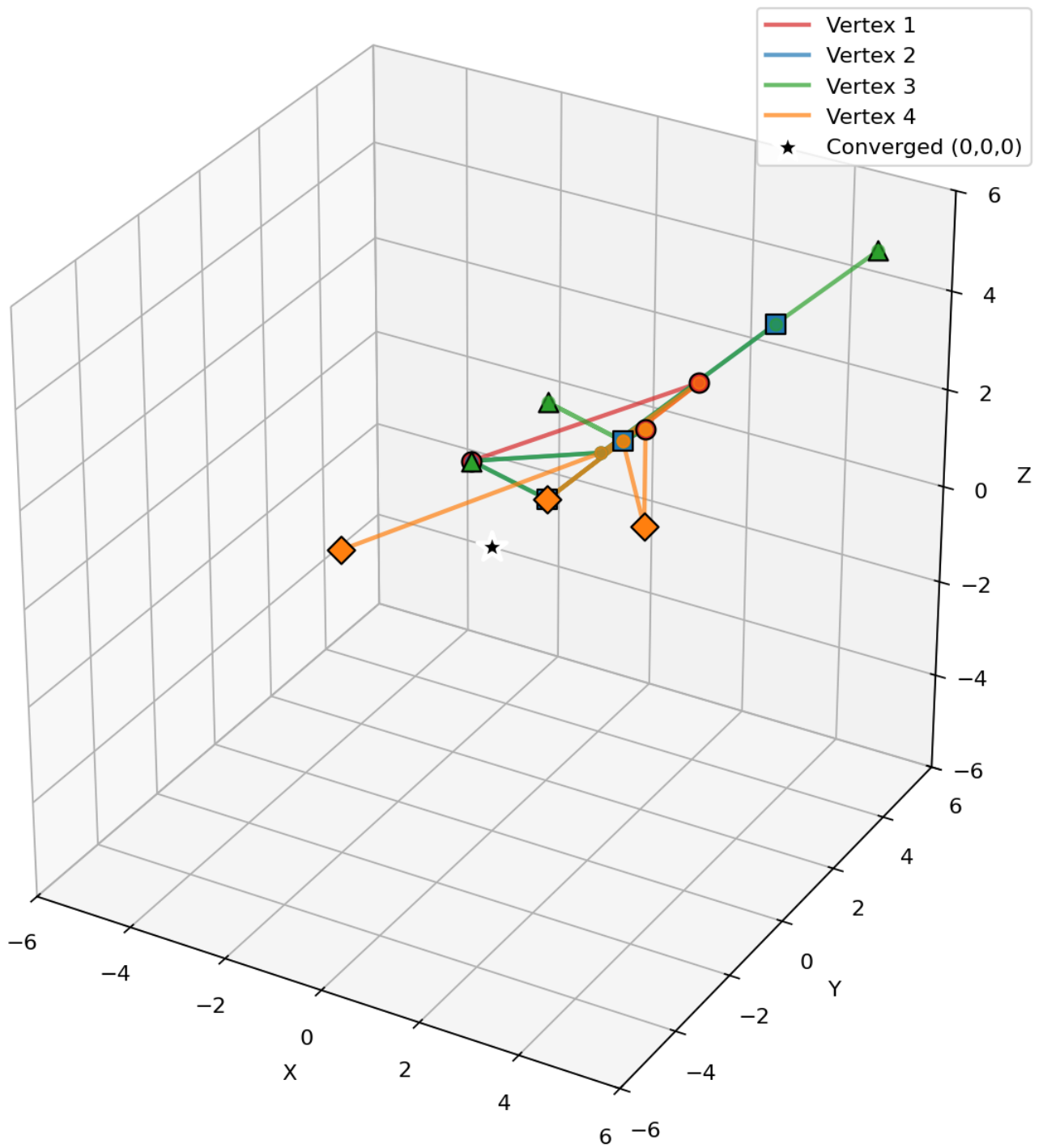
Iteration 6
Best: 3.5, Spread: 1.3



Iteration 9
Best: 0.6, Spread: 3.8



Complete Simplex Optimization Trace
Nelder-Mead on Integer Quadray Lattice

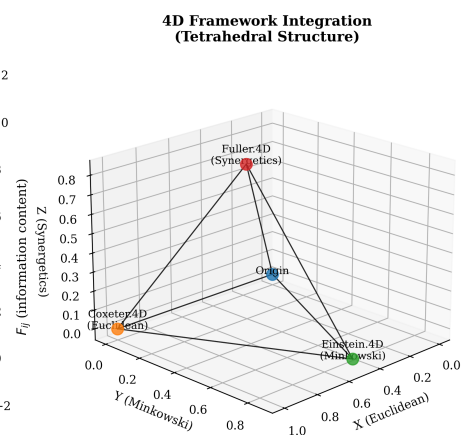
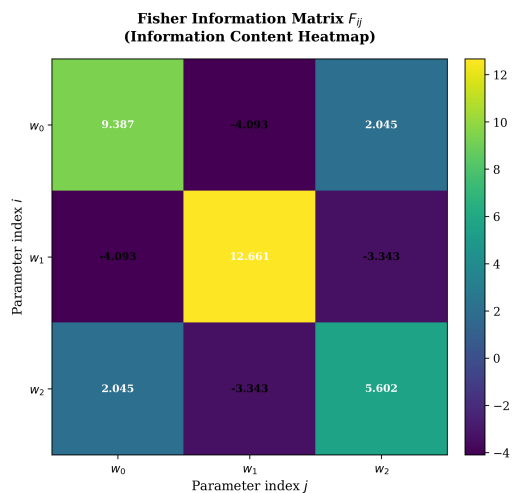
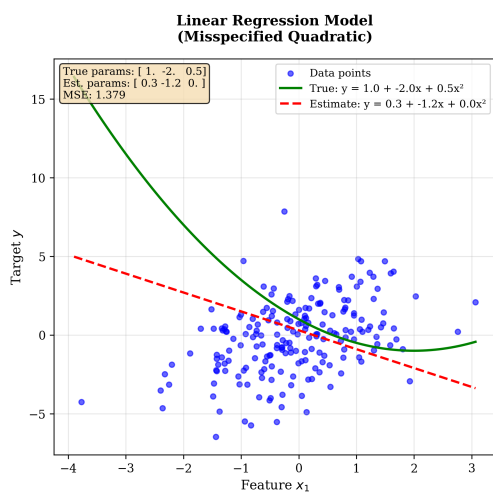


$$\begin{aligned}
& F_{ij} \\
& = (w_0, w_1, w_2)L() \\
& L_n \\
& g_{ij} = F_{ij}\delta_{ij}F^{-1}\nabla L\nabla L
\end{aligned}$$

$$F^{-1}\nabla L$$

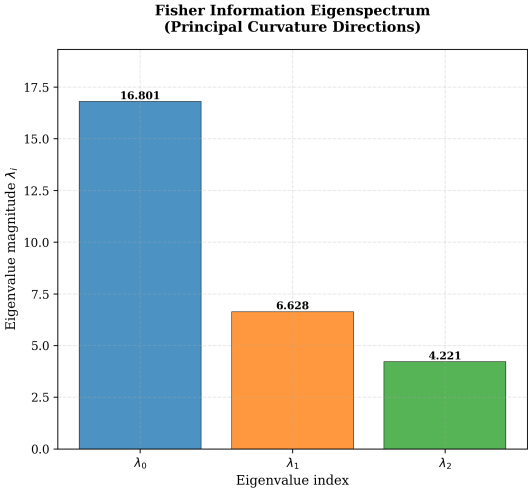
$$\delta_{ij} F_{ij}$$

$$F^{-1} \nabla L \Delta = -\eta F^{-1} \nabla L$$



$$y = w_0 + w_1x + w_2x^2_{true} = [1.0, -2.0, 0.5]w_{est} = [0.3, -1.2, 0.0]F_{ij}F_{ii}w_iF_{ij}w_0, w_1, w_2\delta_{ij}F_{ij}\Delta w = F^{-1}\nabla L$$

 $\mathcal{W}$



Curvature Analysis & 4D Framework

Key Metrics:	
Condition Number:	3.98
Anisotropy Index:	0.592
Total Curvature:	27.651

Eigenvalue Interpretation:

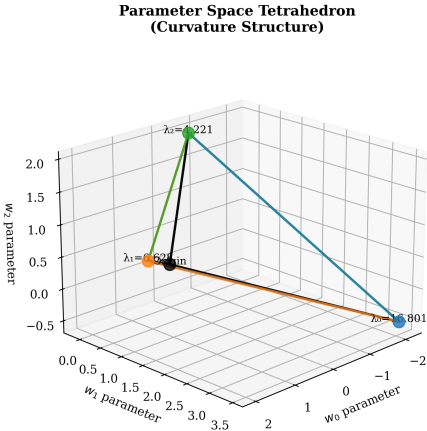
- $\lambda_0 = \therefore 3\%$ Primary direction
- $\lambda_1 = \therefore 3\%$ Secondary direction
- $\lambda_2 = \therefore 3\%$ Tertiary direction

4D Framework Connection:

- Coxeter.4D: Euclidean parameter space
- Einstein.4D: Fisher metric geometry
- Fuller.4D: Tetrahedral structure

Optimization Implications:

- Natural gradient scales by F^{-1}
- Anisotropic scaling improves convergence



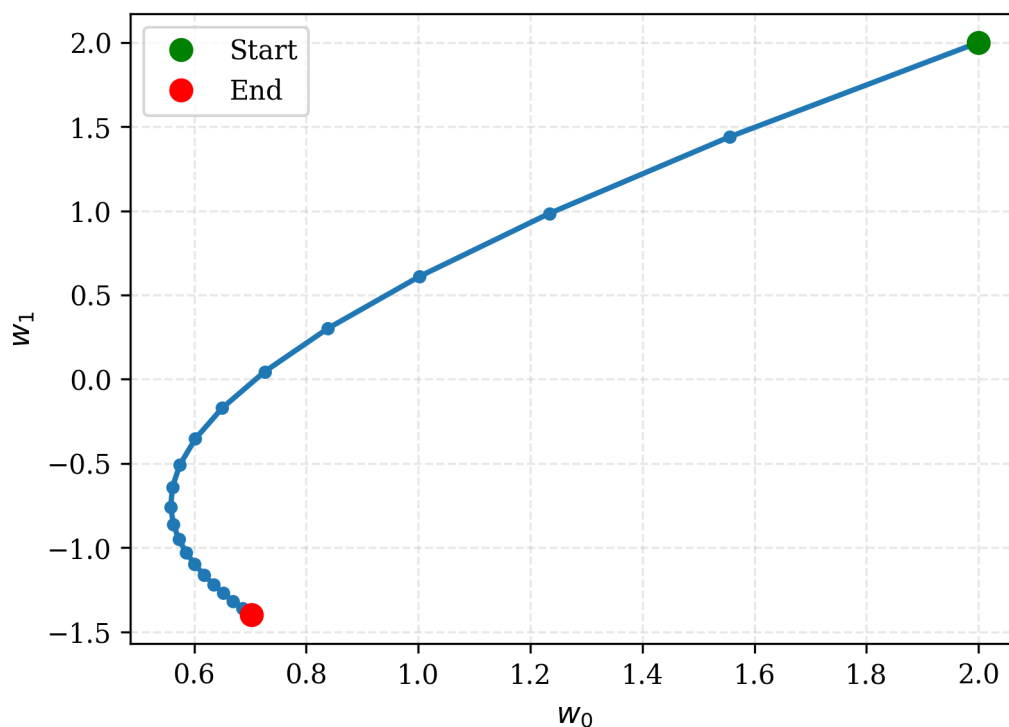
$$\lambda_i F^{-1} F^{-1}$$

$$\eta F \nabla L g_{ij} = F_{ij}$$

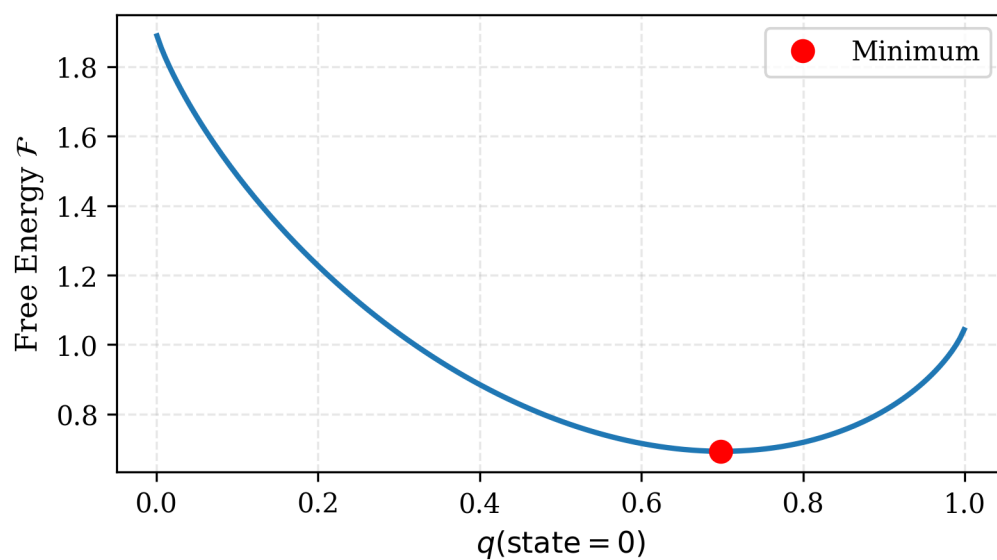
$$F^{-1} \nabla L$$

$$F$$

Natural Gradient Trajectory (Geodesic Motion on Information Manifold)



Variational Free Energy Landscape (2-State System)



$$\mathcal{F} = -P(o|s) + [Q(s)||P(s)]q(=0)\mathcal{F}$$